

Flavio Romano

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Software Engineer and Technical Lead specializing in backend systems, IAM solutions, and cloud-native architectures. Currently pursuing an MSc in Artificial Intelligence while working at NTT DATA, combining enterprise software engineering experience with advanced AI and machine learning expertise.

Experience

Software Engineer / Technical Lead, NTT DATA – Pisa, IT March 2024 – Present

- Contributed to the design and implementation of OAuth 2.0, OpenID Connect, and SAML authentication solutions for a leading Italian telecommunications provider, enhancing a customized Keycloak deployment and enabling adoption of the Phantom Token architecture.
- Developed Spring Boot microservices supporting authorization workflows, including token introspection, redirect management, and JWT/JWE processing.
- Served as Technical Lead for the development of an AI-powered platform automating due diligence documentation for European sustainability regulations, coordinating a team of 4 engineers and leading technical decisions.
- Currently leading a team of 5 engineers in the development of a sustainability platform for estimating CO₂ emissions generated by software systems, hardware infrastructure, and AI workloads.

Software Engineer Intern, NTT DATA – Pisa, IT Dec 2023 – March 2024

- Developed a Java microservice for the GreenIT ecosystem to collect hardware metrics and estimate Software Carbon Intensity (SCI) and CO₂ emissions on Google Cloud Platform infrastructures.
- Designed a configurable monitoring connector that later became the foundation of my Bachelor's thesis.

Technical Skills

Languages: Java, Python, Go, TypeScript, SQL, Bash, C

Backend & Cloud: Spring Boot, FastAPI, Kubernetes, AWS, GCP, RabbitMQ, Redis, Terraform

ML: PyTorch, Scikit-learn, Keras, Pandas, Optuna, RAG, LLM Evaluation

Security: OAuth 2.0, OpenID Connect, SAML, Keycloak

Projects

Compliance Intelligence Platform (codeberg) May 2026 – Present

- Designed and developed an enterprise Retrieval-Augmented Generation (RAG) platform for sustainability compliance and due diligence analysis, enabling grounded question answering over regulatory and corporate documentation.
- Built an end-to-end document intelligence pipeline including text extraction, semantic chunking, embedding generation, metadata enrichment, and vector indexing using Qdrant.
- Implemented an evaluation framework for benchmarking retrieval and generation quality using Precision@K, Recall@K, Mean Reciprocal Rank (MRR), answer relevance, faithfulness, and context utilization metrics.

Encoding Classification using LSTM (github) Apr 2026 – May 2026

- Developed a PyTorch-based sequence classification model to automatically identify the encoding algorithm applied to a text string, distinguishing between Caesar Cipher, ROT13, and MD5 transformations.
- Built and trained a 230k-parameter LSTM model on a dataset of 72,000+ samples, achieving 97.99% test accuracy through token embedding and sequence preprocessing.
- Implemented automated training and inference workflows, enabling real-time prediction of encoding types from arbitrary input strings.

Education

University of Pisa, Master of Science (MSc) in Artificial Intelligence Sept 2025 – Present

University of Pisa, Bachelor of Computer Science Sept 2020 – May 2024